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AI-Based Advancements for Comprehensive Mangrove Analysis Suitability Mapping

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***Abstract*—Habitat loss remains a critical global issue linked to climate change. The Philippines, recognized as a global hotspot for marine biodiversity, is home to extensive mangrove forests. Despite their ecological significance, these forests are increasingly threatened by deforestation and various anthropogenic pressures. To support conservation efforts and mitigate habitat loss, this study employs habitat suitability modeling to identify optimal afforestation zones for mangrove species. Machine learning algorithms, including Maximum Entropy (MaxEnt), Support Vector Machine (SVM), Random Forest (RF), and Artificial Neural Network (ANN), were utilized to analyze mangrove distribution. A total of 17 environmental factors, encompassing bioclimatic and marine variables, along with mangrove sapling presence points, were integrated into the models. Experimental results demonstrated that the RF model outperformed others, achieving an area under the curve (AUC) value of 0.97 in predicting potential mangrove habitats. Based on RF model predictions, the model identified high-suitability zones for mangrove restoration and afforestation. The generated suitability maps provide valuable insights for developing science-based adaptation policies, strategies, and measures aimed at enhancing the resilience of mangrove ecosystems against present and future climate challenges.**

***Index Terms*—distribution, habitat suitability mapping, mangroves, prediction, random forest**

I. Introduction

Habitat loss is one of the global concerns related to climate change. Coastal areas are specifically affected by climate change, resulting in ocean warming, rising sea levels, and a diminished capacity of the ocean to absorb carbon [1]. Protecting natural resources and conserving ecological landscapes are essential strategies for mitigating the adverse impacts of climate change on ecosystems. The use of habitat suitability assessment is a valuable modeling technique that aids in forecasting optimal conditions for plant cultivation, thus helping to prevent habitat destruction and biodiversity degradation [2].

The Philippines is widely considered one of the epicenters of marine biodiversity in the world [3]. The Philippine archipelago is the center of near-shore marine diversity and coastal plants worldwide [4]. Mangroves are an essential part of coastal ecosystems, thriving within the intertidal zone on the landward side of the coastal region. They play a critical ecological role by serving as hatcheries and nursery grounds for juvenile fish, providing shelter and resources before these species migrate to other marine habitats such as coral reefs and seagrass beds. Despite their significance, mangrove forests face a severe problem: mangrove deforestation. The primary driving force behind mangrove forest loss in Southeast Asia and the Philippines is anthropogenic forces that significantly impact the environment [5].

The mangrove forest experienced significant degradation after being converted into aquaculture ponds for decades, leading to the loss of vital ecosystems and disruption of coastal biodiversity [6]. This crisis has been further exacerbated by pollution and human actions, underscoring the urgent need for conservation efforts by the mangrove community. A poignant example of this occurred during Super Typhoon Odette, which struck the Bohol region of the Philippines on December 16, 2021 as shown in Figure 1. The destructive forces of the typhoon, from strong winds and heavy rainfall to storm surges, had immediate and lasting impacts on mangrove communities. The mangrove site in the northwestern part of Bohol was one of the most heavily devastated by Typhoon Odette. With these challenges, there is an urgent need to address the combined effects of global and local stressors that affect mangrove communities.

Recognizing the importance of community awareness in species conservation, this paper proposed a three-year plan to explore patterns and insights regarding marine mangrove characteristics using machine learning techniques. This paper aims to improve the understanding of data behavior, potentially yielding valuable solutions and insights. Effective conservation strategies are based on understanding how species respond to their environment. As such, this paper is subdivided into three parts: (1) assessing the habitat suitability of mangroves in the Philippines, employing data-driven machine learning algorithms such as MaxEnt, SVM, RF, and ANN; (2) creating an automated monitoring approach for mangroves presence detection through the integration of machine learning techniques

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Fig. 1: The mangrove site in the northwestern part of Bohol was one of the heavily devastated due to Typhoon Odette.

and remote sensing; and (3) classifying the mangroves species and real-time mapping location.

II. Related Works

In recent years, there has been a growing recognition of the importance of habitat suitability models in understanding how species are distributed in various environments. Predictive models such as Habitat Suitability Models (SHM), also known as Species Distribution Models (SDM), Environmental Niche Models (ENMs) and Habitat Distribution Models have been utilized to forecast habitat suitability of species by analyzing a range of environmental factors [7] [8]. SDM is one of the ecological tools to study the distribution of a variety of taxa [9]. This model utilizes data on the presence of associated animals to forecast the likelihood of their presence in particular areas, either in different locations or at various points in time.

Diverse investigations concerning mangrove ecology have progressively employed species distribution modeling (SDM) to characterize habitats and project potential habitat suitability for mangroves. SDM employs a computer algorithm to establish a connection between the presence of plant or animal species and the environmental conditions of their habitat, enabling the projection of species-appropriate habitats across geographic regions.

Various research studies have used SDMs as an effective tool for mapping suitable habitats of mangrove species. Recognized as one of the most effective and widely utilized species distribution models, the maximum entropy model (MaxEnt) utilized advanced machine learning algorithms to predict habitat suitability. It utilizes occurrence data and environmental variables to identify optimal habitats for mangrove species across regions. It predicts species distribution using maximum entropy based on limited information about the target species [10]. This model has been widely applied to predict future changes in terrestrial and marine species distributions across various temporal and spatial scales, consistently delivering reliable and accurate predictions [11]. Studies reveal that MaxEnt has been considered effective in predicting the distribution of Rhizophora mangle [12], predictions of suitable habitats for Cerithidea rhizophorarum and Telescopium telescopium, emphasizing temperature as a critical driver, with projections indicating habitat shrinkage due to climate change [13], and an optimized MaxEnt model has been effective in identifying suitable areas for six mangrove species [14].

Random Forest (RF) models have emerged as powerful tools for assessing mangrove habitat suitability and distribution. This model effectively maps mangrove habitat suitability by analyzing environmental variables, achieving high accuracy [15]. This model effectively utilizes non-parametric analytical methods to identify relevant variables, and generate accurate predictions [16].

Support Vector Machine (SVM) has been utilized alongside other algorithms, such as MaxEnt and RF to predict habitat suitability for various mangrove species. In a study on the Sundarban Biosphere Reserve, SVM was part of a suite of ten machine learning algorithms used to assess habitat suitability for 18 mangrove species, highlighting its versatility in ecological modeling [17]. SVM achieved high predictive performance for certain species, indicating its effectiveness in understanding species distribution in response to environmental variables [18]. SVM utilizes only presence data to map input vectors to a higher-dimensional space through kernel functions. As a binary classifier, it is effective in predicting species' habitat suitability and is particularly suitable for analyzing small datasets [19].

Artificial Neural Network (ANN) is one of the machine learning techniques used to assess the mangroves' habitat suitability. This model can process complex datasets, incorporating multiple environmental parameters such as sediment characteristics, elevation, and proximity to water bodies, which are critical for mangrove growth [20]. It also demonstrated versatility in handling diverse ecological data [21]. It can also predict future habitat distributions based on historical data, as seen in the projection of mangrove cover increase [20]. ANN has the ability to adapt its structure through a learning process, enabling it to model nonlinear and complex relationships between input and output data effectively [22].

III. Materials and Methods

A. Study Areas and Mangrove Occurrence Data Collection

The study area covers the northwestern region of Bohol, an island known for its rich biodiversity in the Visayas region of the Philippines. Bohol experiences a tropical climate that supports diverse ecosystems, including extensive mangrove habitats. Mangrove habitat data will be collected through an extensive monitoring campaign conducted along a 31 km coastal stretch encompassing the towns of Calape, Tubigon, Clarin, and Inabanga.

Mangrove occurrence data will be gathered using remote sensing techniques to identify the presence of mangrove habitats, which are classified as presence points. Several mangrove species are found within the area, as illustrated in Figure 1. Five target mangrove species have been identified: *Acrostichum aureum*, (b) *Rhizophora mucronata*, (c) *Lumnitzera*

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TABLE I: Environmental Variables

Type	*Variables*	*Unit*
Bioclimatic	Annual mean temperature	°C
Bioclimatic	Mean diurnal range	°C
Bioclimatic	Isothermality	unitless
Bioclimatic	Temperature seasonality	unitless
Bioclimatic	Maximum temperature for warmest month	°C
Bioclimatic	Minimum temperature for coldest month	°C
Bioclimatic	Annual temperature range	°C
Bioclimatic	Mean temperature of wettest quarter	°C
Bioclimatic	Mean temperature of driest quarter	°C
Bioclimatic	Mean temperature of warmest quarter	°C
Bioclimatic	Mean temperature of coldest quarter	°C
Bioclimatic	Annual precipitation	mm
Marine	Mean sea surface salinity	PSS
Marine	Sea surface salinity range	PSS
Marine	Mean sea surface temperature	°C
Marine	Minimum sea surface temperature	°C
Substrate	Intertidal substrate cover	type

racemosa, (d) *Avicennia rumphiana*, and (e) *Nipa fruticans*, as shown in Figure 2. Furthermore, the World Register of Marine Species (WoRMS) was utilized to verify species names, ensuring accuracy and preventing misspellings [23].

Fig. 2: Target Species Occurencies.

B. Environmental Variables

Mangrove distribution along coastlines is shaped by a combination of terrestrial and marine environmental factors. Key determinants include landscape setting, salinity levels, coastal slope, sea level rise, precipitation patterns, and temperature fluctuations, all of which play a crucial role in influencing mangrove growth and distribution [24].

To model mangrove distribution, a total of 17 ecological variables were utilized, encompassing both bioclimatic and marine environmental factors, as detailed in Table 1. The bioclimatic data set was sourced from WorldClim at a resolution of 2.5 minutes [25], [26], while marine environmental factors were obtained from the Bio-ORACLE v2.2 database with a resolution of 10 km. Several researchers have utilized bioclimatic variables in mangrove habitat suitability models due to their strong connection to climate-related factors, particularly temperature and precipitation [27].

To improve the predictive accuracy of the species distribution models, highly correlated environmental variables were excluded, as strong spatial autocorrelation can negatively affect model reliability. This refinement ensures a more robust and accurate assessment of habitat suitability.

C. Habitat Suitability of Mangrove Under Climate Scenarios Using SDMs

Three stages were involved in generating the predictive maps for the mangroves species distribution such as model fitting, and evaluation, and model prediction. The habitat suitability for the five mangrove species was modeled using the biomod2 package in R Studio, employing five different species distribution models (SDMs). This package integrates machine learning algorithms and statistical models, allowing for the analysis of complex conditions and properties. With this, it can enhance the accuracy of species distribution predictions [28]. In this paper, Maximum Entropy (MaxEnt), Support Vector Machine (SVM), Random Forest (RF), and Artificial Neural Network (ANN) were used.

Species occurrences and environmental variables were used to define the independent variables of the model. Species occurrence data from the study site were used as input for model training and testing. Meanwhile, the environmental data were cropped to the specific boundaries of the study site and rasterized for spatial analysis using min-max normalization to maintain the uniformity of the data, as shown in Figure 3. The environmental and land cover rasters are stacked together. The extracted raster values were compiled into a data frame, forming the basis for training and testing the SDMs.

The prepared dataset is used to train and test habitat suitability models using four advanced machine learning algorithms: MaxEnt, RF, SVM, and ANN. MaxEnt utilizes presence-only data along with environmental variables to predict habitat suitability [29], while RF employs an ensemble learning approach, building multiple decision trees for robust classification [30]. SVM optimizes the separation between species presence and absence points through hyperplane classification, and ANN utilizes deep learning to model relationships between environmental factors and data occurrences.

All SDM models were developed and evaluated using R software, with validation conducted through 10-fold cross-validation, as illustrated in Figure 4. The performance metrics such as Area Under the Curve (AUC), Cohen's Kappa (KAPPA), true skill statistic (TSS), Specificity and Sensitivity were used to validate all the four(4) machine learning techniques. The optimal model was identified by analyzing the Receiver Operating Characteristic (ROC) curve and its associated AUC values. The AUC was computed based on the model's specificity and sensitivity, ensuring a comprehensive evaluation of its predictive performance. An AUC score lower than 0.7 signifies weak discriminative ability, whereas values between 0.7 and 0.8 are classified as satisfactory. AUC values from 0.8 to 0.9 indicate strong performance, while scores above 0.9 reflect excellent predictive accuracy [31].

The habitat suitability maps were created based on the lowest presence threshold, ensuring a clear differentiation between suitable and unsuitable habitats. The probabilities of the presence of species ranged from 0, indicating unsuitability, to 1, representing optimal suitability. A value of 0 indicated no likelihood of presence, intermediate values suggested a

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moderate probability, and a value of 1 represented the highest likelihood of occurrence.

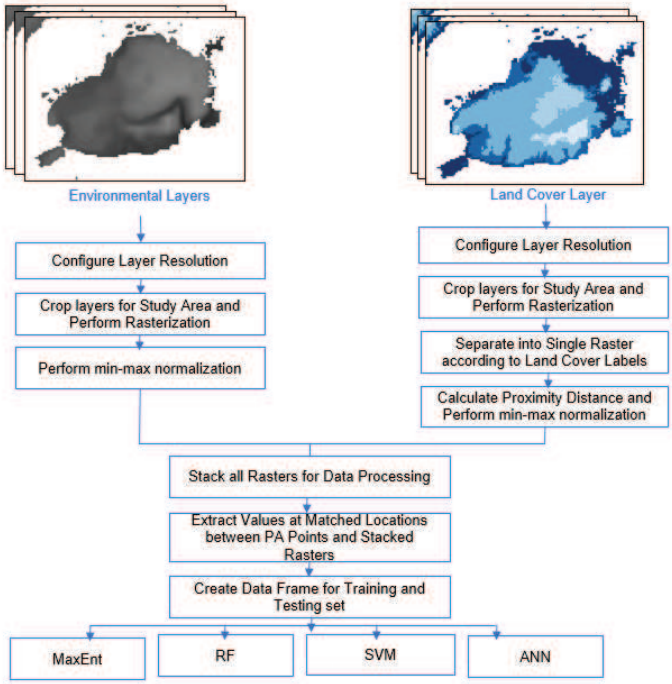

Fig. 3: Habitat Suitability Modelling of Mangrove Species.

The four machine learning techniques were implemented in the R software environment for training, testing, and validation using 10-fold cross-validation to ensure a higher performance assessment, as depicted in Figure 4.

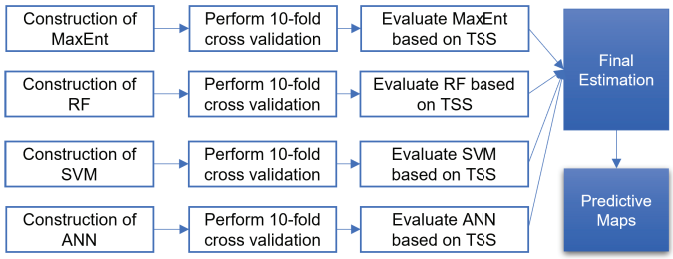

Fig. 4: Crossfold Validation of the SDMs Model

IV. Results and Discussions

A. Model Performance

The model's performance was assessed using the Area Under the Receiver Operating Characteristic Curve (AUC), a key metric for evaluating predictive accuracy. As shown in Table II, the Random Forest (RF) model consistently outperformed other machine learning techniques, achieving a high AUC value of 0.97, indicating its superior ability to distinguish suitable habitats between suitable and unsuitable habitats for *Acrostichum aureum*, (b) *Rhizophora mucronata*, (c) *Lumnitzera racemosa*, (d) *Avicennia rumphiana*, and (e) *Nipa fruticans* across various suitability thresholds. This

TABLE II: SDM Performance

SDM	*AUC*	*Kappa*	Sensitivity	Specificity	TSS
MaxEnt	0.94	0.95	0.94	0.93	0.95
RF	**0.97**	**0.96**	**0.96**	0.96	**0.97**
SVM	0.95	0.94	0.93	0.94	0.95
ANN	0.95	0.95	0.96	0.94	0.95

enhanced performance demonstrates the effectiveness of RF in accurately predicting habitat suitability for these mangrove species. RF's high sensitivity values indicate its reliability in identifying known presence locations, which is crucial for mapping suitable habitats and assessing connectivity. Additionally, its high-resolution suitability maps effectively capture patchy habitat distributions that match the species' known ecology, making them ideal for creating resistance surfaces.

B. Predictive Habitats

Based on the experimental results of the four machine learning techniques used to identify potential habitat suitability, the Random Forest (RF) model demonstrated the highest performance. Consequently, the RF model was selected to generate habitat suitability maps, with suitability values ranging from 0 (low suitability) to 1 (high suitability).

The color gradient on the right indicates the suitability index, ranging from low suitability (light pink) to high suitability (dark green). Areas shaded in green correspond to regions with high habitat suitability for mangroves, while pink and yellow areas indicate moderate to low suitability. The gray regions likely represent areas where mangrove presence is unlikely or where data is unavailable, as illustrated in Figure 5.

Using the data occurrences of mangrove species in the Northwestern part of Bohol and 17 environmental variables, the RF model identifies the habitat distribution of the mangrove. The clustered green patches located in the Central Visayas where the province of Bohol is located suggest optimal mangrove-growing conditions, potentially influenced by tidal patterns, land cover, and proximity to water sources. With this, the distribution of mangroves in the area was shaped by various factors, including bioclimatic conditions, salinity, substrate type, and temperature.

Meanwhile, the dispersed and patchy distribution of high-suitability areas aligns with the known ecological preferences of mangroves, reinforcing the accuracy of the model's predictions.

V. Conclusions

This paper highlights the effectiveness of the Random Forest (RF) model as a powerful tool to map species distributions between mangrove species occurrences and 17 environmental factors. RF is particularly useful for capturing complex species-environment interactions, making it a valuable approach for understanding distribution patterns. The model generates habitat suitability maps for mangrove species, providing crucial information for conservation efforts by helping stakeholders identify priority areas for restoration and protection. Compared to other machine learning techniques such as

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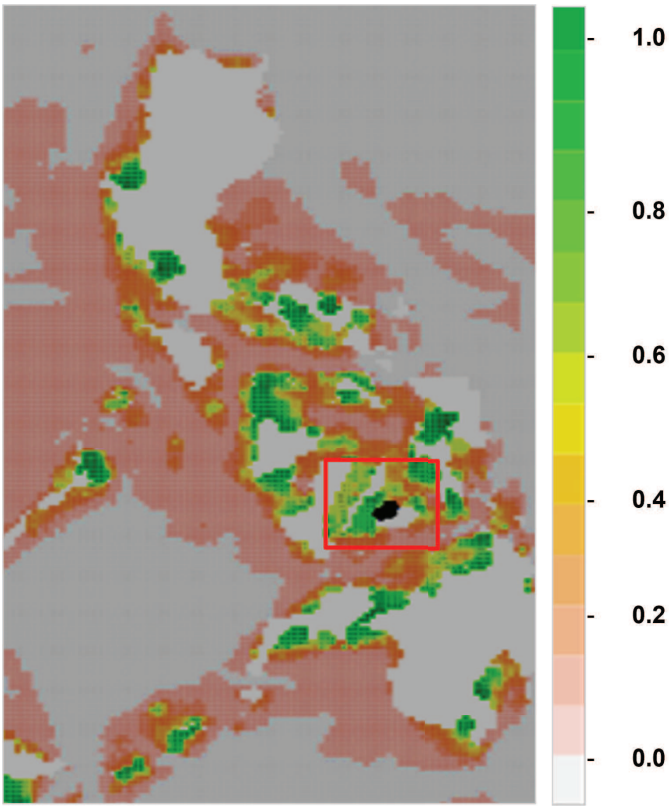

Fig. 5: Predictive Mangrove Habitat Suitability.

MaxEnt, SVM, and ANN, RF excels in identifying suitable habitats by analyzing multiple environmental factors, including climate, terrain, sea surface temperature, and salinity.

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